

Micro-Hydro, If the Stream Allows

Carpathian OER Commons — Utilities Module

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Maturity: L0, a marked draft — a sizing method and a design, published so that they can be read and changed in the open. No stream has been measured, no enquiry sent, and nothing priced beyond the working paper's estimates; the intake screen, the powerhouse, the environmental impact assessment, and the fence are not priced. Read every figure below at this level and no higher. The module moves to L1 when the stream is measured in its driest and its coldest month, the enquiries in Section 10 are answered in writing, and every line in Section 6 carries a dated cost and its basis.

Governed by Section 2 of the compendium. No child appears in this module in any form. The site is not named; site plans are schematic; photographs show structures and systems only. Design loads are site capacity, used for sizing only.

Converted from the energy working paper, EN v0.1, Sections 5 and 7.6. Three things change from the paper, decided on 2026-10-03: the turbine is sized against the energy module's loads at 34 people; the law is set out as found — a special water-use permit, a probable environmental impact assessment, fish protection, and a residual flow, for a turbine of any size; and the public debate on small hydropower in the Carpathians is stated, because a reader should know it before touching a stream.

Permits and local rules. This module is a guide, used in different regions and countries under different laws and guidelines. Depending on where it is built, at what scale, and for whom, permits, registrations, or approvals may be required — for building, water, sanitation, fire, waste, food, or the use of forest and land. Section 10 records what was found for Ukraine, dated, and unchecked until signed; a reader elsewhere checks their own rules with the competent authorities before building, and nothing here replaces that.

1. Purpose

Winter electricity from a stream, if the site has one and the law and the stream allow it: an intake, a buried pipe, a small turbine, and a controller that sends what the site does not use into hot water. It answers the one weakness of the energy module — the darkest month, when the sun is lowest and a mountain stream often runs strongest — and lets the generator rest.

Five rules. The stream first: the turbine takes only what the stream gives above the residual flow agreed in writing, and the reach it pipes keeps running. Measure before anything is priced: head with a level, flow in the driest and the coldest month. Ask before anything is built: the permit, the impact assessment, and the fish come before the turbine. Never stop the water in frost: a pipe that stands still in winter freezes. Nothing taken away: the battery and the generator stay as they are; the turbine adds a source and takes away only part of the array.

Connections. Depends on: a stream within reach; the stream's head and flow, measured; the written answers in Section 10; the energy module, whose circuits it feeds and whose battery and generator it keeps; the market garden's pond, which shares the stream's dry-month flow; the Phase 1 walk, with the Water Code's strip measured. Feeds: the energy module's essential and general circuits; the wood heat module's buffer tank, through the dump load, as an option.

Patterns. Grows from: not yet named. Whether the valley's streams were ever set to work, and how a stream is held here — who may take from it and what is owed to it — are questions for the elders before any intake is built. Feeds: not yet named. Resonates with: 104 Site Repair — the powerhouse outside the strip, on poorer ground; 230 Radiant Heat — a turbine's surplus, through its dump load and the buffer tank, warming rooms rather than wasted. In tension with: 64 Pools and Streams — a turbine takes a reach of the stream into a pipe; the residual flow is what keeps that reach running and in view. 25 Access to Water — an intake and a powerhouse set structures in the belt the pattern keeps open. 71 Still Water — the intake pool is fenced, never a place to swim. The walk status of each is set after the Phase 1 walk, on a later date; until then, unknown (compendium Section 26).

2. Context of validity

Condition	Range this design holds for
Stream	within reach, flowing in its driest and coldest month
Head	about 10 m or more over the pipe's length, [set]
Flow	above the residual flow and the pond's share, [set]
Output	about 0.25 to 1 kW, continuous
Climate	mountain; frost and ice in the intake, Section 3.4
Law	a permit and the enquiries answered, Section 10
Group size	capacity: 34 people, through the energy module

Where the stream is a brook too small to spare a residual flow, or the answers refuse it, there is no turbine, and the energy module stands alone.

3. Sizing method

What you must measure yourself before using this module:

- **Your head** — the height from the intake to the turbine, with a surveyor's level and staff, or a water-filled hose and a pressure gauge, about 0.1 bar a metre. Not with a cheap altimeter, which is commonly out by 46 m or more (Kindberg, Micro-Hydro Power, NCAT, 2011).
- **Your flow** — with a bucket and a stopwatch where all the water can be caught, with a temporary weir where it cannot; monthly for a year if you can, and at least in the driest month and the coldest. A float gives a rough figure only; the sources' correction factors differ, 0.60 to 0.85 in ESHA's guide (2004) and lower in NCAT's.
- **Your pipe's route** — its length and fall, from the walk.
- **Your loads** — from the energy module, Section 3.

Our example: turbine_efficiency, 0.5 to 0.6, water to wire; december_sun and the energy module's loads at capacity, 34 people; exchange_rate, 52 UAH to the euro. Values from the site parameters file, held in the workshop.

3.1 The power

Power in kilowatts is head in metres $\times$ flow in litres a second $\times 9.81 \div 1,000 \times$ the efficiency; at 0.5, about head $\times$ flow $\times 0.005$, as the paper has it.

Head	Flow	At 0.5	At 0.6	A day, at 0.5
10 m	10 L/s	0.49 kW	0.59 kW	11.8 kWh
20 m	10 L/s	0.98 kW	1.18 kW	23.5 kWh
20 m	5 L/s	0.49 kW	0.59 kW	11.8 kWh
30 m	15 L/s	2.21 kW	2.65 kW	53.0 kWh

The efficiency is the whole chain, water to wire. Pelton and Turgo wheels are 80 to 95 percent efficient on their own, crossflows 65 to 85, generators about 80; whole systems 50 to 70 percent (Kindberg, NCAT, 2011); about 60 percent below 50 kW (GTZ, Micro Hydro Power Scout Guide, 2009). Pelton and Turgo hold their efficiency at part flow better than most (Paish, Renewable and Sustainable Energy Reviews 6, 2002).

3.2 What the site needs

Quantity	Calculation	Our example
Darkest-month deficit	$5.5\text{--}6.0 \text{ kWh} \div 24 \text{ h}$	about 0.25 kW
Both circuits, winter, all day	$9.33 \text{ kWh} \div 24 \text{ h}$	about 0.39 kW
Head $\times$ flow for 0.4 kW, at 0.5	$0.4 \div 0.0049$	about 82 m·L/s
For example	$20 \text{ m} \times 4.1 \text{ L/s}$	0.4 kW
Water through it, a day	$4.1 \text{ L/s} \times 86,400 \text{ s}$	about 354 m ³

A set of about 0.4 to 0.5 kW, running all day, carries both circuits through the winter. Anything above that goes to the dump load. A turbine of 1 kW at 30 m of head needs about 6.8 L a second, about 590 m³ a day.

What it lets the array give up. The paper cuts the array from 6 to about 4 kWp when a stream is developed, returning about 30,000 UAH; four fewer 550 W modules at the paper's 8,800 UAH are 35,200 UAH before their share of the mounting. The stream's own driest month is often in late summer, when the sun is strongest, so the smaller array still carries the summer. The battery keeps its four days, because a turbine stops for ice, for drought, and for repair; the generator stays, as a reserve that hardly runs.

3.3 The stream's share

Share	Who decides	Our example
Residual flow, left in the stream	the enquiry, in writing	[set]
The market garden's pond	the market garden module	about 0.08 L/s
The turbine	what is left, in the month it runs	[set]

The residual flow. Ukraine has no approved method for the flow a diversion must leave in a stream; practice follows an older norm, the lowest average monthly flow at 95 percent probability (Huliaieva and Khorev, WWF-Ukraine, 2024). The European reference is the Common Implementation Strategy's Guidance No. 31 on ecological flows (European Commission, 2015). A site cannot compute the norm from a year of bucket readings; the enquiry asks the water agency to set it, and the turbine is sized from what is left.

The pond. The market garden's pond needs about 5 L a minute in a dry week, about 0.08 L a second, from the same stream.

3.4 The intake, the pipe, and winter

The intake. A Coanda screen — a fine-slotted plate the water runs over, which drops the water through and sheds leaves and stones — with slots of about 1 to 1.5 mm, needs no power and clears itself; at one site where basket screens clogged with autumn leaves in hours, Coanda screens raised the year's output by 15 percent (Palmer, International Water Power and Dam Construction, 2004). A screen of this kind passes about 8 to 17 L a second for a 360 mm width (PowerSpout, product data). A screen that fine is also how, on our reading, the intake meets the law's call for a fish-protection device (Section 10).

Ice. Frazil — ice crystals that form in supercooled, fast water — clings to screens and racks and can block an intake; mountain sites, cold and swift and hard to reach, favour it (Ettema, Kirkil, and Daly, Cold Regions Science and Technology 55, 2009). A stable ice cover over the intake pool helps, and so does a submerged intake.

The pipe. Buried below the frost line, polyethylene rather than PVC, and never stopped in frost: if the turbine must be cleared, a purge valve is opened first, and if the pressure falls in a hard frost of about -7°C , the intake is closed and the pipe drained (PowerSpout, Cold weather hydro, 2015; Montana Consumer Guide to Micro-hydro, 2016). The paper's pipe is polyethylene, 110 mm, about 100 m, [set] from the route.

The dump load. An electronic load controller holds the turbine's speed by sending what the site does not use into a resistive load, usually rated at the generator's output (Raja Singh and colleagues, Engineering Science and Technology 21, 2018); a water or space heater is the usual load (Montana Guide, 2016). Here, as an option, an immersion element in the wood heat module's buffer tank: the surplus heats water instead of air.

4. Bill of materials

Quantities for the example. Alternatives are named in the design; none is yet known to work here.

Item	Quantity	Alternatives in the design
Turbine set, about 1 kW, controller	1	Turgo or crossflow
Intake weir and settling basin	1	none named
Coanda screen, 1–1.5 mm slots	1	none named
Penstock, PE 110 mm	about 100 m, [set]	none named
Purge valve, pressure gauge	1 each	none named
Dump load, immersion element	1	an air heater

Item	Quantity	Alternatives in the design
Powerhouse, small, lockable	1	none named
Fence and gate at the intake pool	1	none named
Cable to the energy module	[set]	none named

The siting, as designed. None measured yet; every rule below is checked on the ground and with the authorities before anything is dug.

What	Rule
The diverted reach	as short as the head allows
The residual flow	left at the intake, in every month
The intake	screened, fenced, reachable in winter
The pipe	buried below the frost line
The powerhouse	outside the strip, if the enquiry asks it
The outflow	back into the same stream
Every part	out of the spring's catchment

The Water Code's protective strip is 25 m along a small stream, 50 m on a slope, and building in it is not allowed except for hydrotechnical and some other structures; whether a powerhouse counts as one is asked (Section 10).

The layout, as designed. Intake and screen, settling basin, buried pipe down the fall, powerhouse, outflow to the same stream; cable to the energy module's board; the dump load in the buffer tank. Drawn as relationships only; the walk sets the rest.

5. Build

Step	Trade	Time	If done out of order
The elders asked	household	[from L2]	[from L2]
Head and flow measured, a year	household	[from L2]	mis-sized

Step	Trade	Time	If done out of order
Enquiries answered, permit	household	[from L2]	unlawful intake
Impact assessment, if asked	assessor	[from L2]	permit refused
Intake and settling basin	builder	[from L2]	[from L2]
Pipe laid and buried	excavator	[from L2]	frozen pipe
Powerhouse	builder	[from L2]	[from L2]
Turbine, controller, dump load	electrician	[from L2]	overspeed
Connected to the energy module	electrician	[from L2]	[from L2]
Fence and gate	builder	[from L2]	an open pool

6. Cost

Estimates, not quotations: every priced line is the energy paper's own estimate for autumn 2026, marked "paper"; re-pricing from dated listings was tried on 2026-10-03 and could not be done. Place: not a quotation. Exchange rate: exchange_rate, 52 UAH to the euro, the National Bank's mid-September 2026 rate, rounded.

Item	UAH	EUR	Basis
Turbine, generator, controller, 1 kW	90,000	1,731	paper
Intake and settling basin	40,000	769	paper
Penstock, PE 110 mm, about 100 m	25,000	481	paper
Dump load and wiring	10,000	192	paper
Coanda screen	[set]	[set]	not priced
Powerhouse	[set]	[set]	not priced
Fence and gate	[set]	[set]	not priced
Environmental impact assessment	[set]	[set]	not priced
Special water-use permit	0	0	quoted, Diia

Item	UAH	EUR	Basis
Subtotal, priced lines	165,000	3,173	
Less the array cut to about 4 kWp	−30,000	−577	paper

No percentages are taken on a subtotal this incomplete. The permit is issued without charge (Diia, the state's service guide).

Contribution, not in the total. None. The labour is in the cost, paid.

Running cost. Rent for the water passed through the turbine, under the Tax Code, Art. 255: at the rate shown on a secondary copy, 12.95 UAH for 10,000 m³, which may be out of date, 354 m³ a day for a year would be about 170 UAH. The intake cleared, the screen checked, and the bearings serviced: [set] hours, paid.

7. Operation and maintenance

Responsible: one named adult, paid, with a deputy — [from L2].

- **Daily, in frost:** the pipe's pressure read at the gauge; the intake looked at for ice.
- **Weekly:** the screen and the settling basin cleared; the residual flow seen to pass.
- **Autumn:** leaves cleared from the intake more often while they fall.
- **Yearly:** the turbine and controller serviced; the permit's terms checked against what was taken.

How the programme relates to this work is described in the programme modules, in standing format.

8. Failure modes

From the design, not from experience.

Failure	Early sign	Repair and cost
Screen clogged with leaves	output falls	clear it; [from L2]
Frazil on the screen	output falls in frost	clear; a cover of ice
Pipe frozen	pressure falls	drain it; never stop it
Stream below its share	turbine starves	stop it; the residual first
Controller fails	turbine overspeeds	shut the valve

Failure	Early sign	Repair and cost
Flood at the intake	stones, debris	clear; [from L2]

9. What we got wrong

Nothing yet: this module has not been built.

10. Regulatory notes

Country	Instrument, number, date	Checked by, date
Ukraine	found 2026-10-03, not checked — see below	—
Romania	not checked	—
Poland	not checked	—
Slovakia	not checked	—
Hungary	not checked	—
Serbia	not checked	—
Czechia	not checked	—

Ukraine, found and not checked. Read on 2026-10-03 on secondary copies — protocol.ua, kodeksy.com.ua, and the state's service guide — because the statute site could not be reached; each is read on zakon.rada.gov.ua and signed before this row counts as checked.

- **Water Code, No. 213/95-BP**, Art. 48: special water use is taking or using water from water bodies with structures or devices, energy among its purposes; passing water through hydro structures is excluded from it, except hydropower — so a turbine is special water use. Taking up to 5 m³ a day is excluded, and a turbine takes far more. The state's service guide reads the hydropower exclusion the other way; the Code's own wording is followed until the enquiry answers. Art. 49: special water use needs a permit, issued by the territorial bodies of the state water agency within thirty days. Art. 66: hydropower users keep to the operating rules and provide passage for fish to their spawning grounds. Art. 80: on small rivers, no channel is blocked without passages. Art. 87: water protection zones, set by projects. Arts. 88 and 89: a protective strip of 25 m along a small stream, doubled where the slope is over 3°, in which building is not allowed except for hydrotechnical and some other structures.

- **Cabinet Resolution No. 321 of 13.03.2002**, the permit procedure for special water use, amended by No. 648 of 04.06.2025, not read: a first permit for three years; applications in person, by post, or online; no fee (Diia).
- **Law No. 2059-VIII of 23.05.2017, on environmental impact assessment**, Art. 3, part 3, item 4: hydroelectric power plants on rivers, regardless of capacity, are in the second category, which needs an assessment. Whether a brook counts as a river is asked.
- **Law No. 3677-VI of 08.07.2011, on fisheries**, Art. 17: no water intake may be operated without a fish-protection device.
- **Tax Code**, Art. 255: rent on the water passed through turbines.
- **Order No. 26 of the Ministry of Ecology, 26.01.2017**, on water balances, defines a minimum ecological flow, as reported by WWF-Ukraine; not read.

The fish. Grayling is listed as vulnerable in the Red Data Book of Ukraine, threatened in Carpathian rivers by hydropower and the breaking of rivers into pieces (Mruk and colleagues, Journal for Nature Conservation 81, 2024); campaigners name stream trout and the huchen beside it.

The public debate. Small hydropower is contested in the Ukrainian Carpathians. In 2011 an oblast council approved sites for 360 small plants, about 700 MW; a court annulled the decision in 2012, and two other oblast administrations imposed moratoria that spring (Stankevych-Volosianchuk and Luksha, Ekosphaera, 2013); a third planned one in 2015. Conservation groups — Ekosphaera, WWF-Ukraine, the Ukrainian Nature Conservation Group — oppose new plants for what they do to a river's water and fish and for how little they add to the country's energy, small hydropower being about 2 percent of its renewable supply (Stankevych-Volosianchuk, 2025). Those who build them point to power without emissions, income for a village budget, and the country's renewable targets (Divnych, Zbruč, 2013; Mudra, 2018). A set of one kilowatt for one site is not a plant for the grid; it still takes a reach of a stream, and this module holds it to the same law and the same questions.

The written enquiries: to the territorial body of the state water agency, whether the turbine is special water use, what residual flow it must leave, and on what terms a permit is given; to the regional environmental authority, whether the stream counts as a river under the impact assessment law; to the regional fish protection body, what device the intake needs; to the council, the strip, the water protection zone, and whether a powerhouse may stand in the strip. No intake is built before they are answered.

11. Adaptation notes

The design is a seed, not a copy: a different place grows a different form from it.

- **A steeper stream:** more head, less water; a Pelton wheel, a longer pipe, a smaller share of the stream.
- **A flatter stream:** less head, more water; a crossflow, and a larger residual question.

- **A colder valley:** the intake below a stable ice cover, the pipe deeper, the purge valve and gauge watched daily.
 - **A smaller site:** a few hundred watts carries an essential circuit; the dump load may heat one tank.
 - **No stream, or no answer:** no turbine; the energy module's generator carries the winter.
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12. Provenance and licence

- **Written by:** Michel Garand. [set — others as they choose to be named]
 - **From whose knowledge:** the energy working paper, EN v0.1; Paish, 2002; Kindberg, NCAT, 2011; ESHA, 2004; GTZ, 2009; the Montana Consumer Guide to Micro-hydro, 2016; European Commission, Guidance No. 31, 2015; Huliaieva and Khorev, WWF-Ukraine, 2024; Palmer, 2004; PowerSpout's product data and guidance; Ettema, Kirkil, and Daly, 2009; Raja Singh and colleagues, 2018; Mruk and colleagues, 2024; Ekosphaera, 2013 and 2025; Zbruč, 2013; the Ukrainian instruments in Section 10, as read on secondary copies.
 - **Traditional knowledge:** none. How the valley held its streams is for the elders.
 - **Text:** CC BY-SA 4.0
 - **Drawings and designs:** [hardware licence — open decision]
 - **Measurement data:** none yet
 - **Gate 1:** household decision, 2026-10-03, recorded by Michel Garand — v0.1 may be published as it stands, at L0 as a marked draft, in English, Ukrainian, and German
 - **Wartime review:** 2026-10-03, Michel Garand — v0.1 read against compendium Section 2; nothing places the site; the shelter, the safety link, the battery, the generator, and the fuel store in standing format; accepted
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13. What to check

- **The stream** — whether there is one in reach; its head; its flow, monthly for a year, at least in the driest and the coldest month.
- **The enquiries** — the permit, the residual flow, the impact assessment, the fish-protection device, the strip.
- **The elders** — how the valley holds its streams.
- **The route** — the pipe's length and fall, the powerhouse's place.
- **The efficiency** — the set's own figure, from its maker, against 0.5 to 0.6.
- **Quotations** — every line in Section 6, and the unpriced ones.
- **The rent** — the current rate in the Tax Code.

- **The Ukrainian row** — every instrument read and signed.
 - **From L2, once built:** build times and out-of-order consequences in Section 5; the maintenance record in Section 7; early signs and repair costs in Section 8; Section 9.
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